

TRANSPORT PHENOMENA I

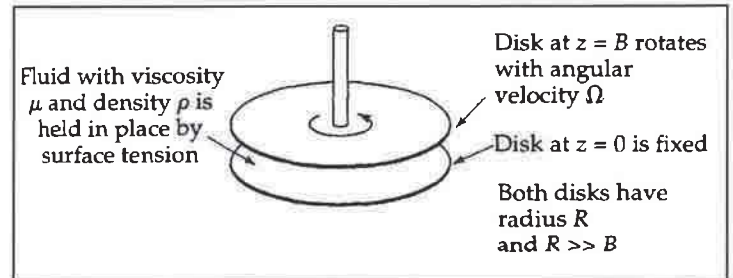
FALL 2025

EXAM 2

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PARALLEL-DISK VISCOMETER

A fluid is placed in the gap between two horizontal circular disks of radius R . Disks are separated by a small distance B . The lower disk is held fixed. The upper disk is rotating at a constant angular velocity Ω .



Postulates:

- Newtonian fluid with constant density ρ and constant viscosity μ .
- $v_r = 0$ and $v_z = 0$.
- Pressure in the fluid is not a function of θ .
- Creeping flow assumption: The flow is very slow and the acceleration term $\frac{D(v)}{Dt}$ in the equation of motion is negligible.

- 20/20 1. Using the continuity equation, prove v_θ is not a function of θ at steady-state conditions.
- 20/20 2. Using the proper equation of motion at steady-state conditions, write down and simplify a partial differential equation for v_θ as a function of r and z .
- 20/20 3. Assuming v_θ may be written as r multiplied by a function of z ,

$$v_\theta(r, z) = r f(z)$$
, convert the partial differential equation for v_θ to an ordinary differential equation for f .
- 20/20 4. Knowing $v_\theta(r, z = 0) = 0$ and $v_\theta(r, z = B) = r \Omega$, write down two boundary conditions for f , solve the ordinary differential equation derived in part 1.3. and evaluate the integration constants.
- 10/20 5. One can estimate the viscosity of the fluid by measuring the torque T_z required to rotate the upper disk at a constant angular velocity. Using the velocity profile derived in part 1.4., find an equation for the viscosity in terms of T_z, Ω, B and R . Hint: torque exerted by the liquid on unit area of the upper disk is the product of the radius r (lever arm) and viscous stress in θ -direction.

§B.1 NEWTON'S LAW OF VISCOSITY

$$[\tau = -\mu(\nabla \mathbf{v} + (\nabla \mathbf{v})^T) + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v})\delta]$$

Cylindrical coordinates (r, θ, z) :

$$\tau_{rr} = -\mu \left[2 \frac{\partial v_r}{\partial r} \right] + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v}) \quad (\text{B.1-8})^a$$

$$\tau_{\theta\theta} = -\mu \left[2 \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) \right] + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v}) \quad (\text{B.1-9})^a$$

$$\tau_{zz} = -\mu \left[2 \frac{\partial v_z}{\partial z} \right] + (\frac{2}{3}\mu - \kappa)(\nabla \cdot \mathbf{v}) \quad (\text{B.1-10})^a$$

$$\tau_{r\theta} = \tau_{\theta r} = -\mu \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right] \quad (\text{B.1-11})$$

$$\tau_{\theta z} = \tau_{z\theta} = -\mu \left[\frac{1}{r} \frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \right] \quad (\text{B.1-12})$$

$$\tau_{zr} = \tau_{rz} = -\mu \left[\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right] \quad (\text{B.1-13})$$

in which

$$(\nabla \cdot \mathbf{v}) = \frac{1}{r} \frac{\partial}{\partial r} (rv_r) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \quad (\text{B.1-14})$$

^a When the fluid is assumed to have constant density, the term containing $(\nabla \cdot \mathbf{v})$ may be omitted. For monatomic gases at low density, the dilatational viscosity κ is zero.

§B.4 THE EQUATION OF CONTINUITY^a

$$[\partial \rho / \partial t + (\nabla \cdot \rho \mathbf{v}) = 0]$$

Cylindrical coordinates (r, θ, z) :

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial}{\partial z} (\rho v_z) = 0 \quad (\text{B.4-2})$$

§B.6 EQUATION OF MOTION FOR A NEWTONIAN FLUID WITH CONSTANT ρ AND μ

$$[\rho D\mathbf{v}/Dt = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g}]$$

Cylindrical coordinates (r, θ, z) :

$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (rv_r) \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] + \rho g_r \quad (\text{B.6-4})$$

$$\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (rv_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta \quad (\text{B.6-5})$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z \quad (\text{B.6-6})$$

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$$1) \frac{\partial f}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (f r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (f v_\theta) + \frac{\partial}{\partial z} (f v_z) = 0$$

$$f = \text{constant}$$

$$v_r = v_z = 0$$

$$\Rightarrow \frac{1}{r} \frac{\partial}{\partial \theta} (f v_\theta) = 0$$

$$\Rightarrow \frac{\partial v_\theta}{\partial \theta} = 0 \Rightarrow v_\theta \text{ is not dependent on } \theta$$

2) θ -momentum balance at S.S

$$\Rightarrow v_\theta = v_\theta(r, z)$$

$$0 = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{\partial^2 v_\theta}{\partial z^2} \right]$$

$$\frac{\partial^2 v_\theta}{\partial z^2} + \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) = 0$$

Next page please

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$$\text{Let } v_\theta(r, z) = r f(z)$$

$$\Rightarrow 0 = \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r \cdot r f(z)) \right) + \frac{\partial^2}{\partial z^2} (r f(z))$$

$$0 = f(z) \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r^2) \right) + r \frac{\partial^2 f(z)}{\partial z^2}$$

$$\frac{\partial}{\partial r} \left(\frac{1}{r} 2r \right) = \frac{\partial}{\partial r} (2) = 0$$

$$\Rightarrow \frac{\partial^2 f(z)}{\partial z^2} = 0 \quad \text{which} \quad \frac{d^2 f(z)}{dz^2} = 0$$

$$\text{BC} \begin{cases} v_\theta(r, z=0) = 0 \\ v_\theta(r, z=B) = r\Omega \end{cases} \rightarrow \begin{cases} f(z=0) = 0 \\ f(z=B) = \Omega \end{cases}$$

$$\Rightarrow \int d^2 f = \int 0 dz^2$$

$$\rightarrow \int df = \int C_1 dz$$

$$\rightarrow f = C_1 z + C_2$$

$$\text{BC} \rightarrow \begin{cases} 0 = C_1 \cdot 0 + C_2 \rightarrow C_2 = 0 \\ \Omega = C_1 B + C_2 \end{cases}$$

$$\Rightarrow C_1 = \frac{\Omega}{B}$$

$$\Rightarrow f = \frac{\Omega}{B} z \Rightarrow v_\theta = r \frac{\Omega}{B} z$$

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We want the viscous force in θ -direction acting on surface perpendicular to z -direction

$$\Rightarrow \tau_{z\theta} = \tau_{\theta z} = -\mu \left[\frac{\partial v_{\theta}}{\partial z} \right] = -\mu \frac{\partial}{\partial z} \left(\frac{r\Omega}{B} z \right) = -\mu \frac{r\Omega}{B}$$

$$\tau_{z\theta} \Big|_{r=R} = -\mu \frac{R\Omega}{B}$$

Lever Arm $\uparrow$ Force/Area $\uparrow$ Area $\uparrow$

We know T_z (Torque) = $R \cdot (\tau_{z\theta} \Big|_{r=R}) \times \pi R^2$

$$T_z = \mu \frac{R^4 \Omega}{B} \pi$$

$$\Rightarrow \mu = \frac{T_z B}{R^4 \Omega \pi} \rightarrow \text{viscosity equation}$$

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Page 6 of 6